

Nikhilesh Vijay Chirutala

AI/ML Solutions Engineer

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SUMMARY

AI/ML Solutions Engineer with 2.8+ years of experience building production software and integrating AI capabilities into application workflows. Strong experience with Python, FastAPI, REST APIs, third-party integrations, AWS, MongoDB, and modern web applications. Hands-on experience with LLM integration, RAG, vector search, and prompt engineering, including building an AI-powered conversational solution for contextual information retrieval. Experienced in API development, integration testing, troubleshooting, documentation-oriented development, and deploying reliable solutions in production environments.

TECHNICAL SKILLS

AI & LLM: RAG, LLM Integration, Vector Search, Prompt Engineering

Languages: Python, JavaScript (ES6+), TypeScript, SQL, HTML5, CSS3

APIs & Integrations: REST APIs, API Development, Twilio, GitHub API, Postman

Backend: FastAPI, Node.js, Express.js, Backend Automation, Authentication, RBAC, ACL

Cloud & DevOps: AWS, Amazon S3, Vercel, GitHub Actions, CI/CD, Linux

Data & Databases: MongoDB, Neon DB, SQL

Frontend: React.js, Next.js, Angular, Tailwind CSS, Bootstrap, Jinja2

Testing & Tools: Postman, Jest, Unit Testing, Git, GitHub

WORK EXPERIENCE

Software Engineer • CA One Tech Cloud Inc.

Apr 2026 – Present | Bengaluru, India

- Develop production backend services and automation using Python and FastAPI, supporting application workflows, APIs, data processing, and cloud-based operations.
- Design and maintain REST APIs and backend integrations, translating business requirements into reliable application functionality.
- Develop AWS-based solutions for application data and storage workflows, with validation, error handling, and operational troubleshooting.
- Investigate production issues, perform root-cause analysis, implement fixes and enhancements, and validate changes through testing before release.

Key Projects:

CONVOSO — Cloud Contact Center Platform

Python, FastAPI, AWS, Amazon S3, REST APIs

- Developed backend services supporting lead routing, call recording management, data processing, and cloud storage workflows.
- Designed automation for recording recovery and synchronization, processing 10,000+ recordings per month.
- Implemented S3 storage validation and metadata synchronization to maintain consistency between application records and cloud objects.
- Added validation, error handling, and troubleshooting workflows for failed operations and unexpected storage states.

Software Engineer • Mantra Technologies

Feb 2024 – Mar 2026 | Hyderabad, India

- Built Python/FastAPI backend services and REST APIs supporting application workflows and third-party integrations.
- Developed and integrated an AI-powered trading application with an LLM/RAG-based conversational experience for contextual information retrieval.
- Designed API-driven application workflows, authentication, RBAC/ACL, and integrations with services including Twilio and GitHub API.
- Tested APIs and end-to-end workflows, resolved production issues, and supported CI/CD and cloud deployments across AWS and Vercel.

Key Projects:

AI Bull — AI-Powered Trading Platform

React.js, Next.js, FastAPI, Python, MongoDB, RAG, LLM

- Developed an AI-powered trading and investment analysis platform for stock and mutual fund analysis, screening, and back-testing.
- Implemented a Retrieval-Augmented Generation (RAG) chatbot using contextual retrieval to generate relevant application responses.
- Integrated LLM capabilities into application workflows, enabling conversational access to relevant financial information.
- Built FastAPI/Python APIs and business logic supporting AI and financial data-processing workflows.
- Tested and refined API/application workflows to support reliable integration of the AI capability into the production application.

CRM.io — AI-Driven CRM System

FastAPI, Python, Jinja2, MongoDB, ACL, Twilio, REST APIs

- Developed CRM modules for Contacts, Tasks, communication workflows, and telephony operations using FastAPI and Python.
- Implemented RBAC and ACL authorization to control access to CRM resources based on user permissions.
- Integrated Twilio for telephony and communication functionality and developed server-rendered interfaces using Jinja2.
- Tested APIs and workflows with Postman and resolved bugs across CRM functionality and production workflows.

GharPe — Real Estate Management Platform

React.js, FastAPI, Python, MongoDB, REST APIs

- Developed the React.js-based admin system for managing real estate projects and controlling their visibility and availability on the website.
- Built project management workflows allowing administrators to create, update, configure, and manage projects displayed across the platform.
- Developed lead management functionality to track and manage customer inquiries generated through the platform.
- Built builder management features for maintaining builder information and associating builders with their respective projects.
- Developed and integrated FastAPI/Python REST APIs supporting administrative workflows, project configuration, and data operations.

PERSONAL PROJECTS

[JS Studio](#) — Online JavaScript Playground

Next.js, GitHub API, Vercel

- Developed an online JavaScript playground for writing and executing JavaScript through a web interface.
- Integrated GitHub APIs and deployed the application with Next.js and Vercel.

[AnyScrum](#) — Project Management Platform

Next.js, Neon DB, Clerk, Vercel

- Developed a project management platform for projects, tasks, and team workflows.
- Implemented authentication and user management with Clerk and persistent application data with Neon DB.

[Guider](#) — Interactive Learning Portal

Angular 19, TypeScript, SSR, Vercel

- Built an interactive learning platform using Angular 19, TypeScript, and server-side rendering with reusable frontend components.
- Deployed the application on Vercel with a production-ready frontend architecture.

[Contact Management App](#) — Contacts Storage Application

React.js, JavaScript, Netlify

- Developed a React-based application for creating, viewing, updating, and managing contact information.
- Built reusable UI components and deployed the application on Netlify.

CERTIFICATIONS

- Agile Methodology Virtual Experience Program — Cognizant (Oct 2022)
- Python Programming Online Training — APSSDC (May–Jun 2021)

EDUCATION

B.Tech — Electrical & Electronics Engineering

2019 – 2023 | GPA: 7.73 / 10

Srinivasa Ramanujan Institute of Technology, Anantapur